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Practical estimation of erosion damage caused by solid particle impact Part 1: Effects of impact parameters on a predictive equation

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Abstract

A practical, predictive equation for estimating erosion damage caused by solid particle impact, which can be utilized under any impact conditions and for any type of material, is proposed. Impact parameters, which affect erosion mechanisms and damage to materials are the size, shape and properties of particles in addition to the important parameters of impact velocity and angle. The material parameters are mechanical properties such as material hardness. Correlative equations with erosion damage are generally derived from particle impact energy, but cannot be applied to the selection of suitable materials in practical service or for estimating erosion damage to newly developed materials. However, another correlative equation derived from indentation behavior, which relates to material hardness, is useful for this purpose of the estimation. The mechanical properties of a material can be therefore regarded as a key parameter for estimating erosion damage. The effects of impact parameters on the correlative equation were investigated in detail for several aluminum, copper, carbon steel and stainless steel specimens. The impact angle dependence of erosion damage to these materials was also discussed. It was concluded that material hardness was clearly an essential parameter and should be a dependent variable in terms of impact velocity dependence and impact angle dependence in the practical, predictive equations.
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Keywords: Estimation of erosion damage; Predictive equation; Solid particle impact; Impact parameters; Mechanical properties; Metallic materials

1. Introduction

Correlative equations between impact parameters and erosion damage caused by solid particle impact have been proposed by many investigators [1–4] but the development of a practical, predictive equation for the actual degradation of materials has not yet been achieved. A predictive equation that can be applied to a variety of types of material and impact conditions is difficult to derive because of the many effective impact parameters that are associated with erosion damage to materials. The impact parameters that affect erosion damage are mainly impact angle, velocity, size, shape and properties of the particles under consideration. The mechanical properties of a material are also a predominant parameter that affects erosion mechanisms. The material hardness is often representative of mechanical properties. On the other hand, erosion damage can widely vary depending on impact angle and the properties of the material. It is generally thought that both material properties and particle impact angle affect the ratio of cutting action or repeated plastic deformation in total material removal. Many attempts have been made to relate one of the above impact parameters to the erosion damage of a target material by fixing the other conditions. As a result, the estimation of erosion damage to a material cannot be necessarily applied to other materials and other impact conditions. The complicated mechanisms or dependent variables of impact parameters make predictive equations for estimating erosion damage more difficult.

The aim of this paper is to investigate the effects of additional factors on erosion damage which depends upon an impact parameter to be noted and to propose predictive equations for erosion damage caused by solid particle impact that can be applied to many types of metallic materials under

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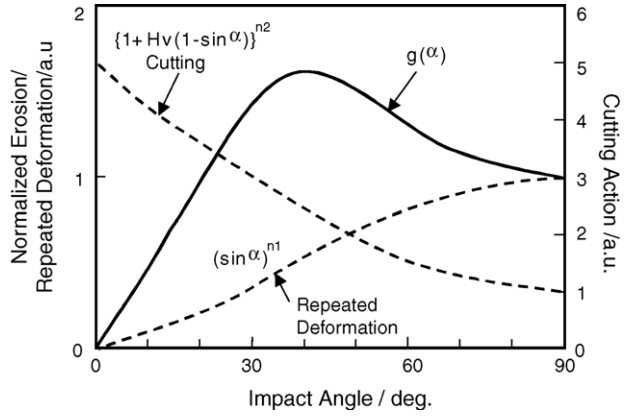


Fig. 1. The concept of erosion arising from repeated plastic deformation and cutting action.

various conditions involving impact angle, velocity, size and the properties of the particles. Erosion tests were carried out under a wide range of the impact conditions, including impact angle, impact velocity and particle size, three types of particles, using not only commercial pure iron, aluminum, but also industrial materials such as carbon and stainless steels. The correlations between the impact parameters and erosion damage at normal angle based on the proposed equations were applied and the results are discussed.

2. Basic equations for predicting erosion damage

In this study, we assumed that the effective parameters for erosion damage include impact velocity, angle, size and type of particles (particle property) and material hardness as one of mechanical properties of a target material. Of these parameters, both impact angle and material hardness seem to be more effective, but difficult to assess in predictive equations because of the variety of impact angle dependence of erosion damage with hardness. In order to discuss the impact angle dependence of erosion damage, we adopted $g(\alpha)$, the ratio of erosion damage at arbitrary angles, to that at normal angle E_{90}. Erosion damage at arbitrary angles $E(\alpha)$ can then be expressed in Eq. (1). $E(\alpha)$ and E_{90} denote a unit of material volume removed per mass of particles ($mm^3\,kg^{-1}$).

$$E(\alpha) = g(\alpha)E_{90} \tag{1}$$

$g(\alpha)$ denotes the impact angle dependence of normalized erosion expressed by the two trigonometric functions and by initial material hardness number Hv in unit of GPa, as shown in Eq. (2):

$$g(\alpha) = (\sin\alpha)^{n_1}(1 + \mathrm{Hv}(1 - \sin\alpha))^{n_2} \tag{2}$$

where n_1 and n_2 are exponents determined by the material hardness and other impact conditions such as particle properties, which include particle shape. $g(\alpha)$ has been defined in the previous paper [5] and is rearranged in this paper, and is conceptually explained in Fig. 1. The first term in Eq. (2) is associated with repeated plastic deformation or the brittle characteristics and with the vertical component of particle impact energy when $n_1 = 2$. The second term shows cutting action, which is relative and more effective at shallower impact angles. The relative cutting action at an arbitrary angle to that at normal angle takes a maximum value at 0°. $g(\alpha)$ is expressed by the product of the two factors in terms of mutual actions. $g(0) = 0$ denotes mixed actions with the greatest possibility of cutting action and impossibility of plastic deformation. In fact, it has been confirmed that $g(\alpha)$ is a useful term, which reflects complicated erosion mechanisms as a function of the type of material [6]. This concept is supported by measurements of plastic strain around indentations [7].

It is well known that erosion damage is related to impact velocity v ($m\,s^{-1}$), particle diameter D (μm), and material hardness within a small range of hardness values or the same compositional systems. E_{90} is initially expressed as a function of impact velocity in Eq. (3):

$$E_{90} = Cv^n \tag{3}$$

the exponent n is typically about 2–3 for metallic materials [4], suggesting that erosion damage is basically proportional to impact energy. A deviation from 2 for the exponent n suggests that other impact conditions or material properties have an effect. The constant C is an arbitrary unit and includes the other functions and the effects of impact parameters. As a result, E_{90} can be expanded as follows:

$$E_{90} = K(\mathrm{Hv})^{k_1}(v)^{k_2}(D)^{k_3} \tag{4}$$

where k_1, k_2 and k_3 are exponent factors, which are affected by other parameters, respectively. K denotes a particle property factor such as particle shape (angularity) and particle hardness, which has no correlation among different types of particles and other factors. It is considered that K denotes an arbitrary unit and an individual value of the particle as particle characteristics. The term of $K(\mathrm{Hv})^{k_1}$ will be mentioned in the next paper in this series [8].

3. Experimental details

A sand blast type erosion test rig was used in this study [9]. This rig was composed of a reservoir tank, valve, acceleration tube and a specimen holder. The particle velocity was controlled by the pressure of a reservoir tank and was different from the air flow velocity in the acceleration tube. The particle velocity depended on the particle size and density under the same tank pressure [8]. The impact angle was controlled from 5 to 90°. Three types of particles, angular SiO_2-1, SiC and glass beads (GB) with mean particle diameters from 49 to 428 μm were used to investigate effects of impact factors on erosion damage. A different lot of SiO_2-2 was used to examine the particle diameter dependence of erosion. The impact conditions are listed in Table 1. These materials used in this study were aluminum, iron–carbon steels and stainless

Table 1
Impact conditions

Particle	Density (kg m^{-3})	Hv (GPa)	Mean size (μm)	Impact velocity (m s^{-1})
SiO_2-1	2600	ca. 20	49	110, 122, 150
			326	74, 91, 104, 121
			428	50, 70, 98, 104
SiO_2-2	2600	ca. 20	49	110
			194	101
			254	108
			326	104
			428	98
SiC	3200	ca. 30	49	108, 146, 167
			254	105
			326	71, 99, 116
GB	2600	ca. 6	200	59, 100, 133

steels, as shown in Table 2. Hv denotes the Vicker's hardness number in units of GPa both in Tables 1 and 2. Copper was used in the next paper in this series [8]. The mass loss of a specimen versus mass of erodent curves were obtained from erosion tests, and erosion rates $E(\alpha)$ or E_{90} (mm^3 kg^{-1}) at steady states were then calculated with the mass loss divided by the density of the materials.

4. Results

Fig. 2(a) shows the impact angle dependence of erosion damage $E(\alpha)$ at an impact velocity of 104 m s^{-1} by SiO_2-1 particles of 326 μm, for several types of stainless steels. It is clear that erosion damage varies with impact angle and the specific type of stainless steels. However, Fig. 2(b) shows impact angle dependence of normalized erosion, $g(\alpha)$, which is generally in the order of hardness for the same materials in Fig. 2(a). The plots are experimental and the smooth curves in Fig. 2(a) and (b) are drawn from $g(\alpha)$ for each material determined by n_1, n_2 and the initial material hardness in Table 2 according to Eq. (2).

Table 2
Target materials, density and hardness

Material	Designation	Density (kg m^{-3})	Hv (GPa)
Aluminum as received	ALR	2700	0.40
Aluminum alloy			
3003 as received	Al1	2700	0.37
5052 as received	Al2	2660	0.59
2017 as received	Al3	2750	1.43
Copper as received	CUR	9020	0.80
Iron as received	FER	7870	1.20
Carbon steel (0.25% carbon)	S25C	7700	1.34
Cast iron as received	CIR	7250	3.00
Carbon tool steel			
Annealed	SK4A	7550	2.20
Quenched	SK4Q	7550	8.00
Spring steel (annealed)	SUPA	7940	2.70
Stainless steel			
Austenite as received	304R	7890	2.90
Austenite heat treatment	304H	7890	1.80
Ferrite + austenite	S1	7730	3.82
Austenite	S2	7930	4.41
Martensite	S3	7740	4.46
Martensite	S4	7670	4.87
Martensite	S5	7670	6.50

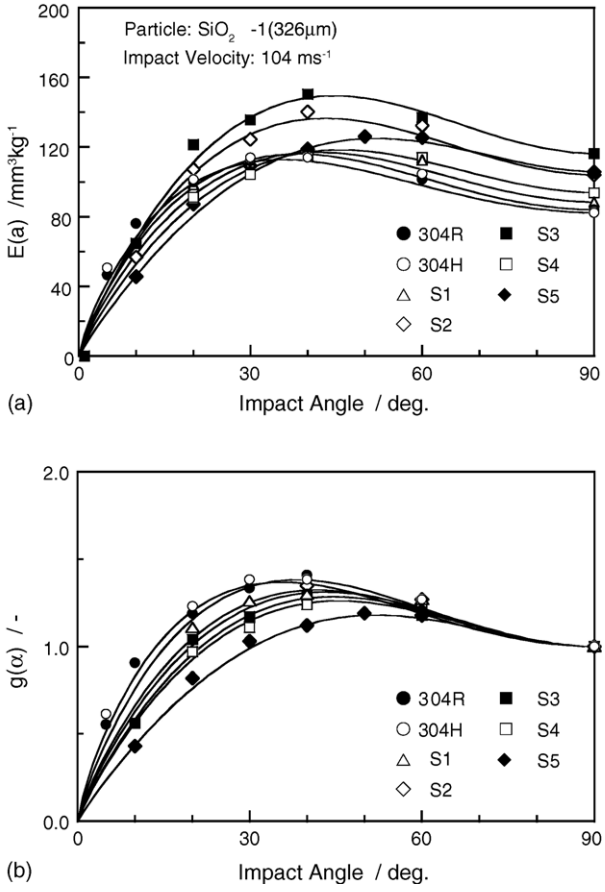


Fig. 2. Impact angle dependence of erosion damage for various stainless steels caused by the impact of silica particles.

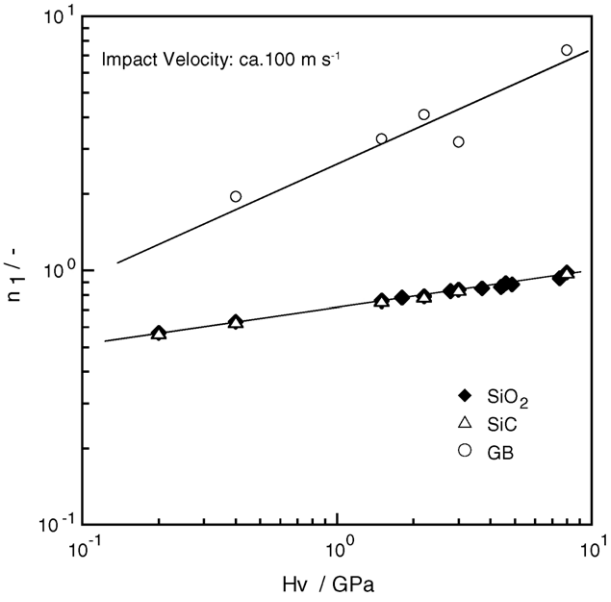


Fig. 3. Relationships between hardness and n_1 for various materials.

Figs. 3 and 4 show the relationships between material hardness Hv and n_1 or n_2, which are plotted for the stainless steel samples, in addition to aluminum, iron and carbon steels [6] on a logarithmic scale. It is found that n_1 and n_2 are related to material hardness despite a wide range of material hardness. The relation of n_1 was nearly the same between SiO_2-1 and SiC particles. The value of n_1 for glass beads (GB) was larger than those for the two angular particles. n_2 shifted from 0.15 to 15 depending on the material hardness for all cases of the particles. A slight difference in the slope of n_2 was observed between the SiO_2-1 and SiC particles, and the slope of n_2 with GB was higher compared to SiO_2-1 and SiC particles.

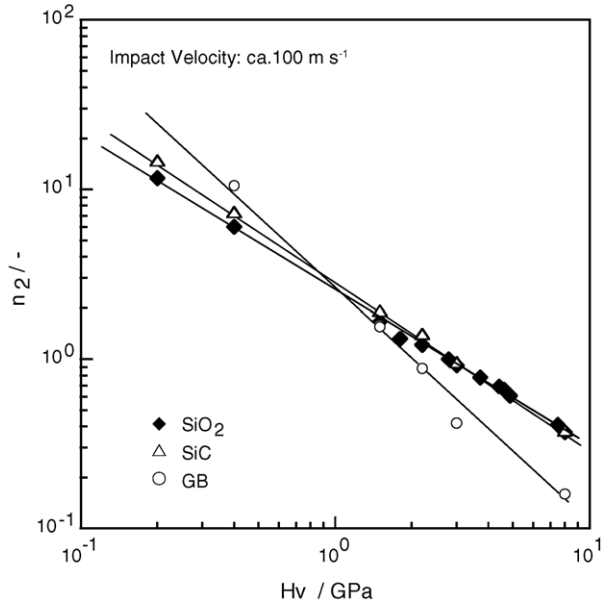


Fig. 4. Relationships between hardness and n_2 for various materials.

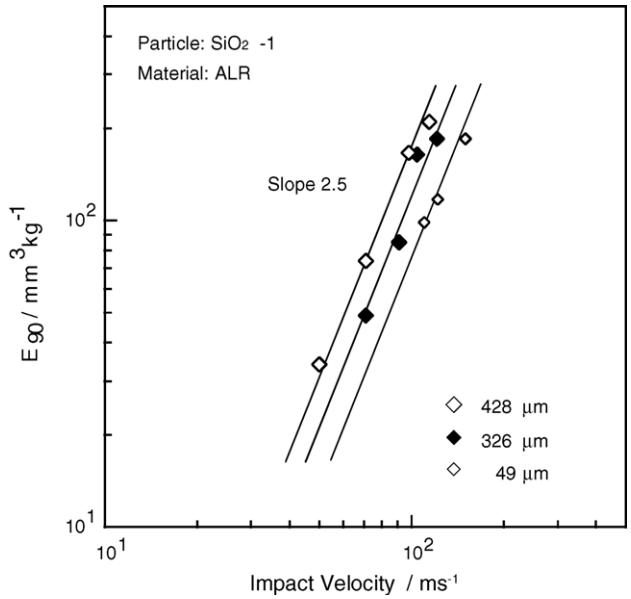


Fig. 5. The effects of impact particle diameter on impact velocity dependence of erosion caused by the impact of silica particles.

Fig. 5 shows an example of the effects of particle size of SiO_2-1 on the impact velocity dependence of E_{90}. The amount of erosion damage by a small particle was lower than that by larger ones for the same impact velocity. The slope of the impact velocity dependence was about 2.5, and was independent of particle diameter for aluminum. Fig. 6 shows the effects of various types of particles on the impact velocity dependence of erosion damage for aluminum (a) and iron (b). It is found that the type of particles seems to change the impact velocity dependence of erosion damage slightly. The slope of impact velocity dependence by SiC was larger than that for other particles both for aluminum and iron, although the amount of erosion damage was similar among the three particles.

Fig. 7 shows examples of the relationships between particle diameter and E_{90} for several materials at an impact velocity of ca. 100 m s^{-1} by SiO_2-2 (a) and SiC particles (b). The impact velocity varied with the particle size at the same tank pressure, as mentioned above and in a previous report [9], and was then modified according to the impact velocity dependence of erosion such as shown in Fig. 6. The slopes of the curves in Fig. 7(a) and (b) were about 0.18–0.22 irrespective of the type of materials and particles on a logarithmic scale. The slope for the other materials had nearly the same values.

5. Discussion

5.1. Effective parameters on the impact angle dependence of normalized erosion

It was verified in a previous paper [6] that both impact velocity and particle size have little effect on the impact angle dependence of normalized erosion for aluminum and iron.

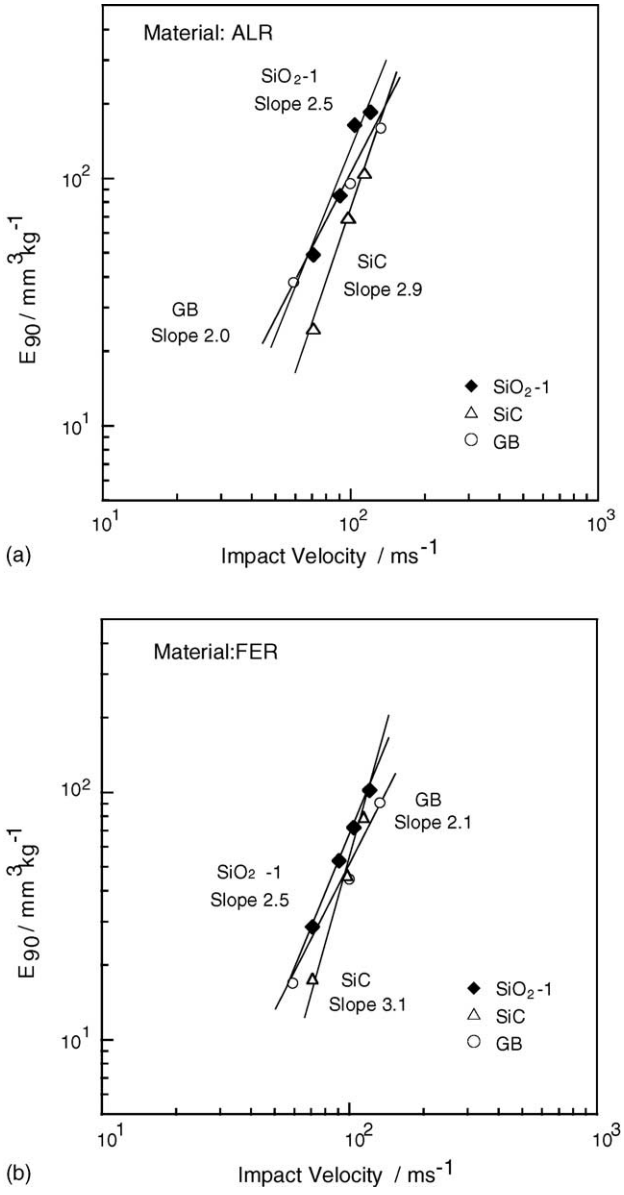


Fig. 6. The effects of various types of particles on impact velocity dependence of erosion for (a) aluminum (ALR) and (b) iron (FER).

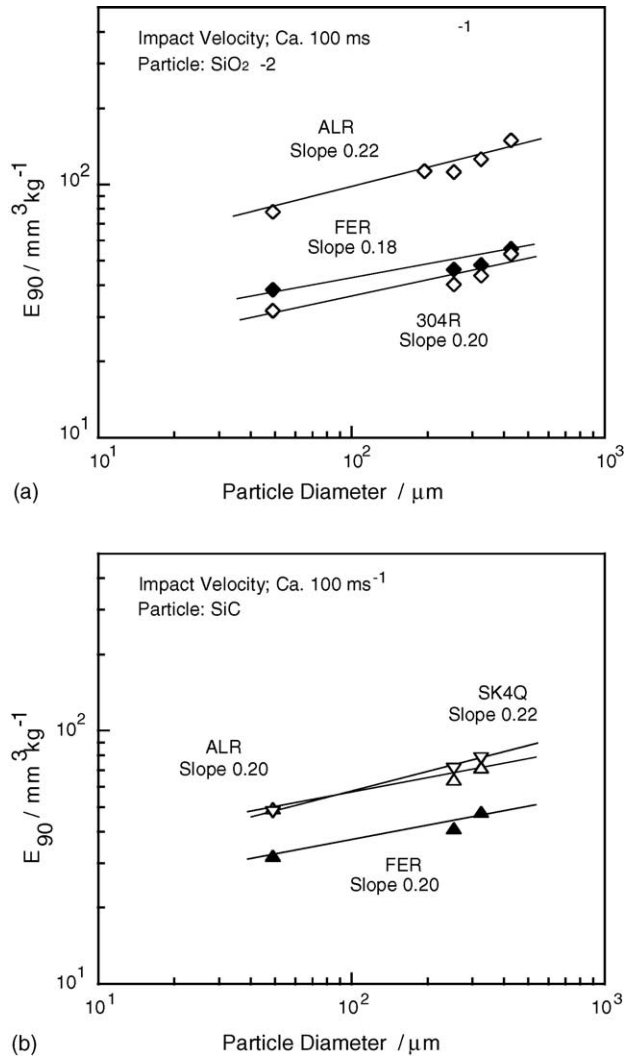


Fig. 7. The effects of various types of materials on particle diameter dependence of erosion caused by the impact of SiO_2-2 particles (a) and SiC particles (b).

The impact angle dependence of normalized erosion for the stainless steel specimens was also stable, with both particle velocity and size. Impact energy is definitely increased both by particle velocity and by size, but the relative aggressiveness of a particle appears to be nearly the same because of constant angularity independent of particle size and the analogy of the penetrated tip configuration of the particle by increased impact velocity. The effects of particle velocity and size on impact angle dependence of normalized erosion is nearly non-existent for these ranges of particle velocity and size.

From Figs. 2 to 4 it is clear that normalized erosion $g(\alpha)$ is directly related to material hardness and the type of particles, which implies particle shape and properties, and that Eq. (2) can be applied to a wide range of industrial materials. Material hardness here denotes initial (non-eroded) hardness. It is well known that well deformed surface hardness is related to erosion damage [10]. However, the surface hardness varies with the erosion conditions [10]. This equation is very useful for the prediction of impact angle dependence of normalized erosion against unknown (untested) materials by adopting the initial hardness. It is especially possible to predict erosion damage at very shallow angles for which experimental data cannot be typically obtained. n_1 and n_2 are considered to show the effects of repeated plastic deformation and cutting action [5]. n_1 for SiO_2 and SiC was the same so that the ability of repeated plastic deformation was similar between the two particles in spite of the different density and angularity. It is probably considered that n_1 is quite the same between SiO_2-1 and SiO_2-2 particles. On the other hand, the value of n_1 for GB was larger than that for angular particles and suggested great repeated plastic deformation (less cutting action, in other words). The high value of n_1 gives a

Table 3
Constants and exponents in n_1, $n_2 = s(\mathrm{Hv})^q$

Particle	n_1		n_2	
	s	q	s	q
SiO_2-1	0.71	0.14	2.4	−0.94
SiC	0.71	0.14	2.8	−1.00
GB	2.8	0.41	2.6	−1.46

functionally lower value in the curve of $(\sin\alpha)^{n_1}$ at low impact angles. For example, the impact angle at the maximum erosion rate is 20° for SiO_2, 30–40° for GB for aluminum, and 40° for SiO_2, 90° for GB for iron [6]. The maximum erosion of a harder stainless steel shifts towards a larger impact angle. These of impact angle dependence of erosion can be precisely obtained from n_1 and n_2 in Figs. 3 and 4. n_1 and n_2 are expressed as a function of initial material hardness, $s(\mathrm{Hv})^q$. s and q are therefore constants which are determined only by the type, shape and property of particles, as the impact angle dependence of normalized erosion is not affected by the other impact parameters such as velocity and size of the impact particles. s and q for the particle properties are listed in Table 3.

5.2. Effects of parameters on impact velocity dependence

The same slopes in Fig. 5 suggest that the impact velocity dependence of erosion is independent of particle diameter and that exponent k_2 does not include the parameter D. Simultaneously, the same slopes in a logarithmic scale imply that the relative E_{90} between the different particle diameters at an arbitrary particle velocity is constant and then the exponent of particle diameter, k_3 is eventually independent of impact velocity.

Both particle properties and the type of materials appear to affect the impact velocity dependence of erosion E_{90} from Fig. 6. Fig. 8 shows the relationships between the material hardness and the exponent of the impact velocity k_2 for the three particles and three types of materials. The value of k_2 is slightly affected both by the material hardness (type of material) and the angular particles. k_2 is constant independent of material hardness in the case of a round particle GB. The difference in k_2 value between SiO_2-1 and SiO_2-2 can be ignored because of the scattered data for SiO_2-1 and the slight difference among the three particles. It is well known that a secondary impact by a fractured particle increases the k_2 value [11]. This is true for one of the accelerated erosion mechanisms, but the fragmentation of stiff SiC particles, which had higher k_2 values, was impossible within the range of impact velocity used, and the particle sizes tested in this study. These results for the value of k_2 were considered to be connected with indentation processes associated with erosion at normal impact, as mentioned in Section 5.4.

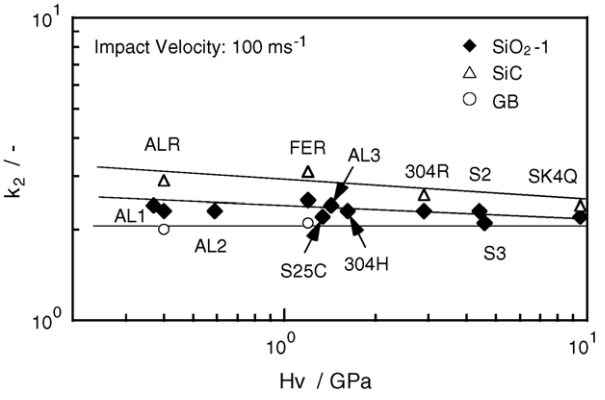


Fig. 8. Relationship between hardness and k_2 for various materials and particles.

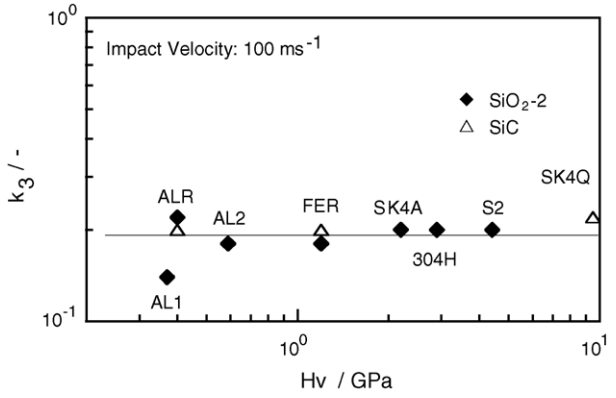


Fig. 9. Relationship between hardness and k_3 for various materials and particles.

5.3. Effects of parameters on particle diameter dependence

It was found that both impact velocity and particle diameter were independent of each other relative to the dependence of E_{90} as mentioned in Section 5.2. The value of E_{90} generally varied with the type of material and particle, but the slopes on a logarithmic scale were very similar (see Fig. 7). Fig. 9 shows the relationship between material hardness and particle diameter exponent k_3. k_3 was scattered within the range of 0.14–0.21 and was negligibly related not only to the type of particles but the material hardness as well. k_3 is therefore regarded to be independent of the other parameters and to roughly take a value of 0.19 in an engineering sense. Considering of the unit of erosion damage ($\mathrm{mm^3\,kg^{-1}}$), the impact energy to the unit mass of erodent should be quite the same, irrespective of particle size and the effects of particle diameter may not necessarily appear (this means $k_3 = 0$). In this case, the deviation in the value of k_3 ($k_3 = 0.19$) from 0 suggests that particle size has only a slight effect and the increase in material removal is caused by the large penetration of a particle tip due to the increase in the particle size.

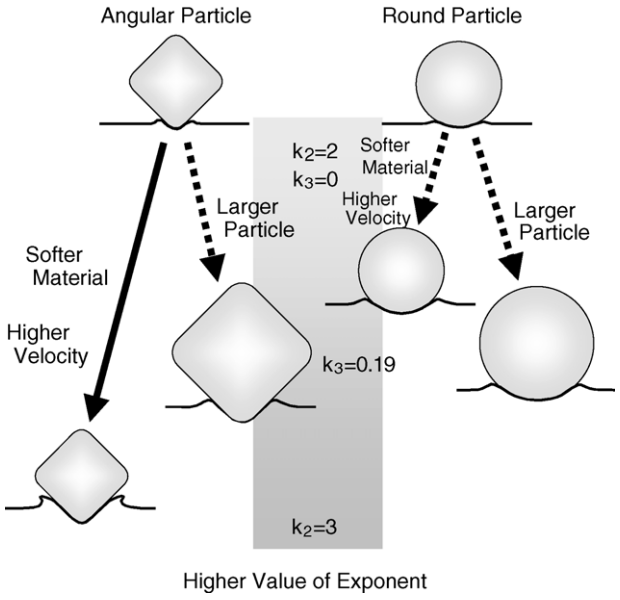


Fig. 10. Model of the effects of impact parameters on exponents k_2 and k_3.

5.4. Principles of the increase in exponent value by the parameters

The increase in impact velocity or particle diameter clearly accelerates erosion damage. From the fact that an increase in particle velocity or size leads to larger or deeper indentations as schematically shown in Fig. 10, deviations in k_2 and k_3 values from the theoretical ones ($k_2 = 2$, $k_3 = 0$) indicate the true effects of impact velocity and particle diameter which are connected with the relative aggressiveness of indentation. The larger or deeper is the indentation the greater amount of material is removed from the rim of the indentation. Taking into account the irregular indentation by angular particle such as SiO_2 and SiC, the more angular particle of SiC would be expected to penetrate more deeply and cause more effective material removal (larger k_2) than that by a round particle GB. The slight decrease in k_2 value with an increase in material hardness in Fig. 8 can be attributed to the decreasing effects of a smaller indentation due to the increased hardness.

6. Conclusions

The effects of alternative factors on erosion damage, which depend upon an impact parameter, were investigated when the impact parameters were selectively fixed to be impact velocity, angle, particle size and property and material hardness. The conclusions are:

1. The impact angle dependence of normalized erosion is useful as basic equations for predicting erosion damage for a wide range of industrial materials and the effective parameters were confirmed to be initial material hardness and particle property. n_1 and n_2 were good indicators for predicting erosion mechanisms associated with repeated plastic deformation and cutting action. In particular, the impact angle dependence for an unknown material can be easily estimated from its initial hardness.
2. Both impact velocity and particle size were independent of each other; that is, particle size did not affect the impact velocity dependence of erosion damage and the impact velocity did not affect the particle diameter dependence of erosion damage.
3. The impact velocity dependence of erosion damage was affected not by particle size but rather, particle properties and material hardness.
4. The deviations in k_2 and k_3 values from the theoretical values can be attributed to the relative aggressiveness of indentation associated with the penetration of the particle tip.

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References

[1] I. Finnie, G.R. Stevick, J.R. Ridgely, The influence of impingement angle on the erosion of ductile metals by angular abrasive particles, Wear 152 (1992) 91.
[2] J.G.A. Bitter, A study of erosion phenomena part I, Wear 6 (1963) 5.
[3] J.G.A. Bitter, A study of erosion phenomena part II, Wear 6 (1963) 169.
[4] G.L. Sheldon, A. Kanhere, An investigation of impingement erosion using single particles, Wear 21 (1972) 195.
[5] Y.I. Oka, H. Ohnogi, T. Hosokawa, M. Matsumura, The impact angle dependence of erosion damage caused by solid particle impact, Wear 203–204 (1997) 573.
[6] Y.I. Oka, M. Nisimura, K. Nagahasi, M. Matumura, Impact angle dependence of erosion by solid particle impact for metallic materials, Zairyo-to-Kankyo 48 (1999) 355–361.
[7] Y.I. Oka, K. Nagahashi, Measurements of plastic strain around indentations caused by the impact of round and angular particles, and the origin of erosion, Wear 254 (2003) 1267.
[8] Y.I. Oka, T. Yoshida, Practical estimation of erosion damage caused by solid particle impact. Part 2: Mechanical properties of materials directly associated with erosion damage, Wear 259 (2005) 102–109.
[9] Y.I. Oka, N. Nishimura, K. Nagahashi, M. Matsumura, Control and evaluation of particle impact conditions in a sand erosion test facility, Wear 250 (2001) 736.
[10] Y.I. Oka, I.M. Hutchings, Dependence of material hardness on erosion by solid particle impact, Boshoku Gijutsu, JSCE 39 (1990) 610.
[11] G.P. Tilly, W. Sage, The interaction of particle and material behaviour in erosion processes, Wear 16 (1970) 447.